MIL-C-27556A (USAF)
12 February 1970
Superseding
MIL-C-27556 (USAF)
3 April 1962

MILITARY SPECIFICATION

CARTRIDGE KIT, PARACHUTE EJECTION MLU-14/B
AND
CARTRIDGE KIT, PARACHUTE EJECTION MLU-15/B

1. SCOPE

1.1 Scope. This specification covers the requirements for two similar cartridge kits, designated as the MLU-14/B (0.00 second delay) and MLU-15/B (0.75 second delay) respectively, hereafter referred to as the Kit. These cartridge kits are intended to be used in ejecting the parachute for retardation of the BDU-12/B Practice Bomb.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-25918	Cartridge Actuated Devices, Aircraft Crew Emergency Escape, General Specification for
MIL-P-116	Preservation, Methods of

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:23Z
Check the source to verify that this is the current version before use.

MILITARY STANDARD DRAWINGS

MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS

Air Force

61D14986	Cartridge Kit, Parachute Ejection MLU-14/B (0.00 second delay).
61D14987	Cartridge Kit, Parachute Ejection MLU-15/B (0.75 second delay).

Army Munitions Command

10522490	Gas Generator XM17
10522498	Initiator, Cartridge Actuated, XM64 Assembly
10522503	Initiator, Cartridge Actuated, Delay, XM86 (with XM169 Cartridge).

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Material shall be selected and processed for the various parts in accordance with applicable specifications and standards as indicated on the drawings.

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:23Z
Check the source to verify that this is the current version before use.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, a suitable standard part (MS or AN) is not available from any of the suppliers who stock MS and AN parts. In this case, the commercial part must be replaceable by the standard part (MS or AN) without alteration, and the corresponding standard part number must be referenced in the parts list.

3.3.2 Use of parts. Except for the use authorized in 3.3.1, AN and MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Components

3.4.1 MLU-14/B (0.00 second delay). The Cartridge Kit, Parachute Ejection MLU-14/B shall consist of the following major components or equal:

	Nomenclature	Drawing No.	Quantity
a.	Initiator, Cartridge Actuated, Delay, XM86 (with XM169 Cartridge)	Army Ordnance Corp 10522503	1
b.	Gas Generator XM17 (with M37 Cartridge)	Army Ordnance Corp 10522490	1

3.4.2 MLU-15/B (0.75 second delay). The Cartridge Kit, Parachute Ejection, MLU-15/B shall consist of the following major components or equal:

	Nomenclature	Drawing No.	Quantity
a.	Initiator, Cartridge Actuated, XM64, Assembly (XM148 Cartridge)	Army Ordnance Corp 10522498	1
b.	Gas Generator XM17 (with M37 Cartridge)	Army Ordnance Corp 10522490	1

3.5 Design and construction

3.5.1 MLU-14/B. The Cartridge Kit, Parachute Ejection MLU-14/B shall be designed and constructed in conformance with drawing number 61D14986 and subsidiary drawings.

3.5.2 MLU-15/B. The Cartridge Kit, Parachute Ejection MLU-15/B shall be designed and constructed in accordance with drawing number 61D14987 and subsidiary drawings.

3.5.3 Torque requirements. The principal components of the kit shall be so constructed that when assembled with set screws installed, the cap-initiator and head-generator body joints will withstand a torque of 400 inch-pounds without separation.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:23Z
Check the source to verify that this is the current version before use.

3.5.4 Residual magnetism. The initiator and generator shall be satisfactorily demagnetized so as to remove any residual magnetism as a result of manufacturing or inspection processes. Each assembled device shall not deflect a pilot's standby compass, type AN-5766 or equivalent, more than three (3) degrees when placed six (6) inches from the face of the compass.

3.5.5 Safety provision. Each initiator shall be provided with ground safety pins for use during shipping, storage, handling and maintenance. With the ground safety pin installed, the firing pin shall not withdraw past the safety pin when subjected to a force of 220 pounds applied along its longitudinal axis.

3.6 Functional requirements

3.6.1 Cocking force. The minimum force required to cock the firing mechanism of either initiator shall be 15 pounds. Not more than 30 pounds of force shall be required to cock the initiators.

3.6.2 Ejection method. Gas pressure generated by the initiator shall be transmitted through the connecting tube to the firing mechanism of the generator. The cartridge in the generator shall provide sufficient energy to meet the ballistic requirements of this specification.

3.6.3 Ballistic requirements. The M37 Cartridge shall generate sufficient pressure in the XM17 generator to shear a 7,200 pound shear joint within 0.15 seconds after initiation of the cartridge and to meet the requirements of 4.7.8.2.1.b.

3.7 Environmental requirements. The Kit shall be capable of meeting all functional requirements when stabilized at any temperature between minus 65°F and plus 160°F.

3.8 Identification of product. The Cartridge Kit, major components, parts and assemblies shall be marked for identification in accordance with MIL-STD-130.

3.8.1 Lot numbering. Each separate production lot of each explosive component shall be assigned an ammunition lot number in accordance with MIL-STD-1168. This lot number shall be marked on each item in addition to identification specified in MIL-STD-130.

3.9 Ammunition data card. Ammunition data cards shall be initiated, distributed and maintained in accordance with MIL-STD-1167.

3.10 Government furnished property. When specified in the contract, the Government will furnish the following to the contractor, upon his request, for use in making the MLU-14/B and MLU-15/B Cartridge Kit:

Source: https://assist.dla.mil -- Downloaded: 20261O-20T12:23Z
Check the source to verify that this is the current version before use.

ITEM NO	STOCK NO	PART NO	ITEM DESCRIPTION	QUANTITY
1	1377-540-4813	8593745	Cartridge, M 37	*
2	1377-050-5973	10522709	Cartridge, Delay, XM148	*
3	1377-050-5974	10522707	Cartridge, Impulse, M169	*

*Quantities supplied shall be sufficient to meet the production and test requirements.

3.11 Workmanship. The Kit, including all parts and accessories, shall be fabricated in a manner that will insure that the criteria of appearance, fit and adherence to specified tolerances are observed. Particular attention shall be given to neatness and accuracy of markings of parts and assemblies. All components shall be free from cracks, splits, cold-shuts, and inclusions. There shall be no burrs, or sharp edges which might mar adjacent equipment or injure personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the Kit shall be as follows:

a. Preproduction Tests (see 6.4a)
b. Acceptance Tests (see 6.4b)

4.3 Preproduction testing

4.3.1 Preproduction test samples. The preproduction test articles shall be representative of production Kits. Location of testing shall be designated by the procuring agency in the contract.

4.3.1.1 Preproduction sample size. The quantity of items to be used for preproduction testing shall be designated in the contract - see 6.2.1.

4.3.2 Preproduction tests. Preproduction tests shall consist of all tests described herein under 4.7 "TEST METHODS."

4.4 Acceptance tests. Acceptance tests shall consist of:

a. Individual Tests
b. Sampling Tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:23Z
Check the source to verify that this is the current version before use.

4.4.1 Individual tests. Each Kit shall be subjected to the following individual tests:

a. Examination of Product (4.7.1)
b. Conformance to Drawings (4.7.2)
c. Residual Magnetism Test (4.7.3)

4.4.2 Sampling tests. Sampling tests of the Kit shall consist of the following:

a. Safety Pin Resistance Test (4.7.5)
b. Ballistic Tests (4.7.8)

4.4.2.1 Sample size. Lot sample size for the Safety Pin Resistance Test shall be made in conformance to MIL-STD-105, Inspection level II. Acceptance number is zero, rejection number is one (1).

4.4.2.2 Ballistic test sample. Test sample sizes for the Ballistic Test shall be in accordance with following table, with the next two lots after a rejected lot being handled as first production lots:

FIRST PRODUCTION LOT

Production Lot Size	Sample Size
1 to 800	15
801 to 1800	21

4.4.2.3 Rejection and retest. When the sample item fails to meet the specified tests, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.4.3 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of this specification.

4.4.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in accepted items. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Safety. Tests involving firing or possible inadvertent firing of kits, initiators, or cartridges shall be performed behind suitable barricades or in proper inclosures.

4.6 Packing, packaging and marking. The inspection shall ascertain that the packing and packaging of the cartridge kits and their markings conform to Section 5 of this specification.

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:23Z
Check the source to verify that this is the current version before use.

4.7 Test methods

4.7.1 Examination of product. Each component shall be given a complete visual examination and inspection to determine compliance with the requirements of this specification, including workmanship. Failure to satisfy the requirements of this specification shall be cause for rejection.

4.7.2 Conformance to drawings. Each component shall be inspected for strict conformance to the engineering drawings, and to all specifications and standards referenced therein. Failure to meet the above criteria shall be cause for rejection.

4.7.3 Residual magnetism test. The Kit, completely assembled, shall be tested for residual magnetism in accordance with MIL-C-25918. Failure to meet the demagnetization requirements of MIL-C-25918 shall be cause for rejection.

4.7.4 Break-away torque test. The Kit, completely assembled, including all set screws, but without explosive components, shall be secured in a fixture so as to preclude damage or deformation of parts or false torque. A torque wrench set at 400 inch-pounds shall be applied to the head-body and cap-body joints. Any movement within the joint or breaking of joint members shall be cause for rejection.

4.7.5 Safety pin resistance test. This test shall be performed on the firing mechanism less explosive components, with the safety pin inserted, apply a load of 220 pounds once in each direction along the longitudinal axis of the firing pin. Failure of the safety pin to prevent movement of the firing pin from the "safe" position shall be cause for rejection.

4.7.6 Cocking force test. With a suitable scale attached to the loop of the firing mechanism, less explosive components, the force required to manually move the firing pin into the "cocked" position shall be measured. A measured force of less than 15 pounds or more than 30 pounds, or failure of the mechanism to remain in the "cocked" position shall be cause for rejection.

4.7.7 Initiator energy test. Two XM86 or XM64 initiators, as applicable, one conditioned at -65°F, and one conditioned at 160°F, shall be tested separately in the following manner:

a. Connect the initiator to a 15 foot length of aircraft hose conforming to MS28741, size 4.
b. Connect a suitable gauge to opposite end of hose.
c. Fire the initiator and measure the generated pressure at the gauge. A measured pressure of less than 500 psi shall be cause for rejection of the lot of initiators.

4.7.8 Ballistic test

Source: https://assist.dla.mil -- Downloaded: 20260-20T12:23Z
Check the source to verify that this is the current version before use.

4.7.8.1 Test set-up. For Ballistic Testing, the system shall be mounted in an approved test fixture. This fixture shall include a 7,200 pound shear joint and an 80 pound weight, positioned to permit unrestricted horizontal movement. Instrumentation shall be provided to measure shear time and velocity.

4.7.8.2 Test procedure. The sample lot shall be divided into three groups, stabilized to temperatures shown below and fired:

Group 1	Group 2	Group 3
-65°F to -70°F	65°F to 75°F	160°F to 165°F

4.7.8.2.1 Test criteria

a. Allowable time from initiation of cartridge in initiator to breaking of 7,200 pound shear joint:

(1) MLU-14/B Kit, 0.25 second maximum.

(2) MLU-15/B Kit, 0.85 second maximum.
0.65 second minimum.

b. Velocity of 80 pound weight. The ejector shall accelerate the 80 pound weight to a minimum velocity of 120 feet per second within a distance of not more than 42 inches.

Failure to meet these time requirements, or travel of more than 42 inches before the 80 pound weight achieves the specified velocity shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation & packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Items will be preserved and packaged in accordance with MIL-P-116 and special instructions received from the procuring activity.

5.1.2 Level C. Items will be packaged in accordance with FED-STD-102.

5.2 Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A Items will be packed in accordance with instructions received from the procuring activity.

5.2.2 Level C. Items will be packed in accordance with FED-STD-102.

5.3 Marking. Marking will be in accordance with MIL-STD-129.

6. NOTES

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:23Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The Cartridge Kit, Parachute Ejection MLU-14/B and MLU-15/B are intended to be used in ejecting the parachute for retardation of the BDU-12/B Practice Bomb.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Conditions under which Government Furnished Property will be made available to the contractor.
c. Whether preproduction tests will be conducted by the procuring activity.
d. Level of preservation packaging and packing. (See 5.1 and 5.2)

6.2.1 Preproduction test samples. It is expected that the contract or purchase order will specify that MLU-14/B or MLU-15/B Cartridge Kits, as applicable, representative of the production cartridge kit will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order, should specify the point of inspection for these tests.

6.3 Production delivery. No production items should be delivered prior to approval of the preproduction samples.

6.4 Definitions

a. Preproduction testing. Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order, to prove, prior to his starting production, that the equipment meets the requirements of this specification.
b. Acceptance tests. Acceptance tests are those tests accomplished on equipment submitted under a contract to determine that the units are equivalent in performance and construction to the approved preproduction samples.

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force

Preparing activity:
Air Force - 70

Project No. 1377-F203

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:23Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Ogden Air Materiel Area
ATTN: OONEO
Hill Air Force Base, Utah 84401

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:23Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 1531

Source: https://assist.dla.mil -- Downloaded: 202610-20T12:23Z
Check the source to verify that this is the current version before use.